

MUTHUKKUMARAN B

Software Development Engineer | Backend Developer | Full-Stack Web Developer | Aspiring AI Engineer

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PROFESSIONAL SUMMARY

Second-year Information Technology student with strong Data Structures & Algorithms fundamentals (LeetCode, HackerRank) and hands-on backend/full-stack development experience in Python/Django. Built a full-stack modular ERP system spanning 14 business domains and two additional production-style web applications, including LLM and computer-vision integrations, plus hands-on teaching experience as a Technical Trainer Intern. Seeking an SDE, Backend Developer, Full-Stack Developer, or AI Engineering internship/role to build scalable, real-world software.

TECHNICAL SKILLS

Languages: Java, Python, C++, C

Frameworks & Libraries: Django, HTMX, Alpine.js, Tailwind CSS, OpenCV, MediaPipe, NumPy

Core Concepts: Object-Oriented Programming, Data Structures & Algorithms, Time & Space Complexity Analysis, REST APIs, CRUD Operations, Unit Testing

Tools & Platforms: Git, GitHub, VS Code, SQLite, Celery, Ollama (local LLM), g++

EXPERIENCE

Technical Trainer Intern — CSC Computer Education, Erode

Apr 2026 – May 2026

- Trained 50+ students in Python, C, and C++ through structured technical sessions and hands-on lab exercises.
- Designed practice problem sets and doubt-clearing sessions that strengthened students' problem-solving fundamentals and coding accuracy.
- Received a certificate of excellence from the organization for communication and technical delivery.

PROJECTS

Mini ERP System

June 2026

Django | HTMX | Alpine.js | Tailwind CSS | SQLite | Celery | Ollama (LLM)

- Built a full-stack, modular ERP web application covering 14 business domains, including sales, purchasing, inventory, manufacturing, procurement, delivery, foreign trade, and blockchain-based supply chain tracking.
- Designed a reactive, AJAX-driven interface using HTMX and Alpine.js on a Django backend, delivering a dynamic UI without a heavy JavaScript framework.
- Integrated a local LLM via Ollama to power an in-app voice assistant, and implemented audit logging, real-time dashboards, and Celery-based background task processing.
- Managed the full development lifecycle in Git/GitHub, including branching strategy and collaborative commit workflow.

Repo: <https://github.com/muthukkumaranb/minierp>

Platter to Purpose

Feb 2026

HTML5 | CSS3 | JavaScript | Python | Django | Git | GitHub

- Built a full-stack community platform connecting users to social resources, support services, and volunteer opportunities.
- Implemented user authentication, database models, dynamic routing, and CRUD operations in Django to manage community resource listings.
- Built a responsive, mobile-first, accessibility-compliant front end using HTML5, CSS3, and JavaScript with cross-browser support.

Repo: https://github.com/muthukkumaranb/platter_to_purpose

Air Canva — Virtual Drawing Application

Dec 2025

Python | MediaPipe | OpenCV | NumPy

- Built a gesture-controlled drawing app that lets users draw on a digital canvas using real-time hand movement captured via webcam.
- Implemented real-time hand tracking and finger-gesture recognition with MediaPipe to control drawing, erasing, and colour selection.
- Used OpenCV for video capture/frame processing and NumPy for pixel-level manipulation, enabling smooth, low-latency rendering.

Repo: <https://github.com/muthukkumaranb/aircanva>

EDUCATION

Bachelor of Technology — Information Technology

2025 – 2029

Sri Krishna College of Engineering and Technology, India | SGPA: 9.05 / 10.0

Relevant Coursework: Data Structures and Algorithms, OOP with C++, Discrete Mathematics, Computer Organization and Architecture, Operating Systems

ACHIEVEMENTS & CERTIFICATIONS

- Solved Data Structures and Algorithms problems on LeetCode, HackerRank, and other competitive platforms, covering arrays, linked lists, trees, graphs, dynamic programming, and sorting.
- Competed in 10+ hackathons; reached the final round at the Odoo Hackathon and the Madras Institute of Technology (MIT) Hackathon.
- Student Hackathon Coordinator, B.Tech Information Technology — Sri Krishna College of Engineering and Technology.